Week 4 — Reading Pack

Curated excerpts from industry white papers, the syllabus readings, and the PMSA items — organized by learning objective

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Required reading before Week 5 · ≈35 minutes (plus the Murphy clustering sections, ≈40 minutes)

Assigned alongside: Murphy, Probabilistic Machine Learning: An Introduction, chapter 21 (Clustering) — §§21.1–21.4; the PMSA webinar PDF "Pharma Marketing: The Upside of Segmentation"; the Segmentation Field Guide (reference).

Why a reading pack. Segmentation vocabulary in pharma is learned from the documents practitioners write, not from the clustering chapter. Part A sets the stakes in the industry's own words (verbatim, with source, from IQVIA papers in the course library). Part B is the practitioner's description of profiling → orchestration — what a segmentation is for. Part C is the data foundation a segmentation stands on. Part D is patient analytics and the governance that comes with it. Part E walks you through the syllabus readings and the PMSA items. Read, answer the reflection questions, bring one answer to Week 5. PDFs are in the "reading assignment" folder for this week.

Learning objectives for Week 4 (from the syllabus)

1. Design a behavioral segmentation that marketers can actually act on.
2. Compare unsupervised (k-means, GMM, hierarchical) vs. supervised (decile, propensity) segmentation.
3. Evaluate segment stability, separation, and commercial interpretability.

Key concepts: decile-based vs. behavioral vs. needs-based segmentation; feature engineering from claims/Rx (writing behavior, patient mix, switching, channel response); persona development and the bridge to creative/messaging strategy.

A. Why the Marketing Director asks the question (objective 1)

Two excerpts from IQVIA's 2025 survey of commercial leaders. The first says targeting and personalization are the highest-priority, lowest-adoption use cases in sales and marketing; the second says why — the data a segmentation needs is siloed. Both are the reason a segmentation project is usually a data project first.

> *Sales and Marketing: High priority, low adoption — Sales and marketing are rated highest in strategic importance, yet low current average adoption. Despite the promise of AI to drive field productivity, omnichannel targeting, and content personalization, many organizations have not yet industrialized these tools. The barriers are less about ambition and more about scaling enablement across decentralized teams.*
>
> — IQVIA, "AI in Life Sciences Commercialization" (2025 survey of 107 senior commercial leaders), AI impact by commercial function

> *While organizations use varied data sources — such as CRM, field activity, claims, prescriptions, digital engagement, patient support, and market data — these are often siloed and inconsistently structured, limiting their utility for enterprise-scale AI. As one respondent put it, "We have the data — we just can't use it the way we need to."*
>
> — IQVIA, "AI in Life Sciences Commercialization," Barriers to scale

Reflection A. Map the data sources in the second excerpt onto tonight's four feature families (writing behavior, patient mix, switching, channel response). Which family depends on a source the brand does not own? Which one depends on a source the brand itself generates — and why is that a problem for a segmentation (hint: notebook §2, calls by volume quintile)?

B. From profiling to orchestration — what a segmentation is for (objectives 1 & 3)

IQVIA's 2026 paper on medical scientific engagement describes, for medical science liaisons, exactly the pipeline tonight built for the commercial field: profile the customer, prioritize, then sequence touchpoints and log outcomes to decide the next best action. Read it as a description of what the segmentation's consumer needs from you.

> *High-quality engagement starts with robust customer profiling and flows into strategic orchestration of personal and non-personal channels. MSLs map influence networks, referral routes, formulary constraints, and evidence needs to prioritise stakeholders with pathway leverage. They then sequence touchpoints to sustain scientific momentum. For example, a focused virtual deep-dive is followed by a compliant, approved email summary and then a peer-to-peer roundtable to progress any evidence gap, with outcomes from each step logged to determine the next best action.*
>
> — IQVIA, "Redefining Global Medical Scientific Engagement" (2026), Insight-driven engagement: From call plans to orchestrated journeys

> *To support more targeted preparation, the pilot used an "expert ecosystem" approach. A method that considers multiple dimensions of expertise, including scientific activity, clinical practice, digital presence, and peer recognition. Looking across these complementary perspectives helped MSLs build a clearer picture of who influences evidence, decision-making, and knowledge flow within their local systems.*
>
> — IQVIA, "Redefining Global Medical Scientific Engagement," same section

> *Structured peer-to-peer engagement and strategic-thinking workshops strengthened scientific dialogue, clarity, and influence, while embedding data-driven planning, customer segmentation, and channel orchestration where scientific application mattered most.*
>
> — IQVIA, "Redefining Global Medical Scientific Engagement," Key findings of the International MSL Pilot

Note three things. First, 'formulary constraints' is on the profiling list — the access feature tonight (reject rate) is not an afterthought in practice. Second, the profile has dimensions (scientific activity, clinical practice, digital presence, peer recognition) that map onto feature families, including one — peer recognition / influence — that tonight's data does not have. Third, 'outcomes from each step logged to determine the next best action' is the exit trigger on every persona card, and Week 5's subject.

Reflection B. Write the four dimensions of the 'expert ecosystem' as features you could build from data a commercial team has (Rx panel, claims, CRM, web, publications/conference data). For each, say whether it is a behavior of the HCP, a decision the brand made, or an attribute of the HCP's environment — and which of the three a segmentation should be built on.

C. The data foundation a segmentation stands on (objective 3)

A segmentation is refreshed, re-scored and consumed by other systems. IQVIA's 2026 piece on enterprise data in the agentic era says what that requires: governed core customer data, and a temporal view of the customer — which is exactly the feature-window / outcome-window discipline from tonight.

> *The answer is not to slow innovation. It is to distinguish between what must be standardized and where teams can adapt and experiment. Core customer, product, territory, consent, compliance, and performance data require disciplined governance. Execution layers, including campaign design, content personalization, channel orchestration, and local experimentation, can then move faster because they are anchored to trusted foundations.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise" (Silvis, 2026)

> *AI agents are expected to reason across data assets. A current master data record may show the latest state of a customer, but many business questions require understanding how that customer changed over time. Without temporal context, semantic clarity, and end-to-end lineage, agents may have access to information but lack the context required to use it responsibly.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise" (Silvis, 2026)

> *As commercial teams move toward more personalized and dynamic engagement, the margin for ambiguity narrows. Omnichannel execution depends on the ability to connect touchpoints across field, digital, service, access, and engagement channels. If those touchpoints are not governed through a common data spine, teams may gain speed in isolated pockets while increasing fragmentation across the enterprise.*
>
> — IQVIA, "AI Alone Will Not Modernize the Life Sciences Enterprise" (Silvis, 2026)

Reflection C. 'How that customer changed over time' is the share-trend feature and the stability test. Describe, in three sentences, the refresh process for tonight's segmentation: what is re-scored monthly, what is re-clustered and how often, and what has to be true (stability ARI, per-HCP agreement) before the field's lists change. Then say which of the 'core customer' fields in the first excerpt your segmentation writes back to.

D. Patient analytics and governance (objective 1, patients)

IQVIA's paper on trustworthy AI/ML for patient analytics is the clearest industry statement that a patient segment is an output about a person and carries obligations. Two excerpts: data minimization, and purpose limitation.

> *Through input transformation, IQVIA ensures that only essential attributes are retained. Features with high reconstruction risk or low modeling utility are excluded during preprocessing. Data is pseudonymized externally and abstracted early, reducing the need for collection of detailed raw inputs. The use of synthetic trends derived from group-level statistical abstractions reflects this principle in practice: models are built using only what is necessary — and nothing more.*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research," Collection limitation and data minimization

> *Non-reversibility reinforces the boundary between raw input and model use. The system is designed so that data collected for AI/ML can only be used for approved purposes. Outputs such as cohort scores or patient segments are labeled with metadata that restricts downstream use to a specific context and time frame. By making outputs unusable for anything but the intended application, non-reversibility enables robust enforcement of purpose limitation.*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research," Use limitation and purpose specification

> *Each of the three architectural strategies is embedded within a broader governance model that supports continuous oversight. Input transformation pipelines are versioned and logged. Risk assessments tied to non-reversibility thresholds are stored and periodically audited. Latent space models are reviewed for drift and abuse potential. These controls establish traceability and institutional accountability.*
>
> — IQVIA, "Trustworthy AI/ML for Patient Analytics and Research," Accountability and oversight

Reflection D. Tonight's patient segmentation produced a 'cost-sensitive strugglers' segment (23% of patients, copay $69, PDC 0.51, 42% discontinuation). Write the metadata label the second excerpt asks for: purpose, permitted downstream use, prohibited use, refresh date, owner. Then name one feature in the patient file you would exclude under the first excerpt's rule ('high reconstruction risk or low modeling utility') and say why.

E. The syllabus readings and the PMSA items — what to look for (objectives 2 & 3)

Murphy, Probabilistic Machine Learning: An Introduction (2022), chapter 21 — Clustering

The chapter is free online (probml.github.io). Read §21.1 (introduction, incl. evaluating clustering outputs), §21.3 (K-means) and §21.4 (mixture models as clustering); skim §21.2 (hierarchical) and §21.5 (spectral). Look for:

- **Evaluating a clustering without labels.** Murphy is explicit that unsupervised evaluation is hard and that purity, Rand index and mutual information all need a reference partition. Tonight we had one (the planted truth); in practice you will not — which is why the stability test (a reference partition you make yourself by resampling) matters.
- **K-means as a special case of a GMM.** Spherical, equal-variance components with hard assignments. That is the assumption the profile heatmap violates (the long tail is a different shape from the core), and why GMM's BIC kept improving to k = 8 tonight.
- **Choosing k.** Murphy's treatment of the elbow, silhouette, and model-selection criteria (BIC for mixtures) — and his warning that the criteria disagree. Tonight they did (silhouette said 3).
- **Initialization and local minima.** K-means++ and multiple restarts. Tonight more restarts found a lower-inertia solution that was worse (the 72-HCP junk cluster) — the objective is not the goal.

PMSA items (links in “External readings — links.md”)

Item	What it is	What to look for
“Pharma Marketing: The Upside of Segmentation” (PMSA webinar PDF, Dec 2020)	End-to-end case: how a real HCP segmentation was built and operationalized	The feature list; how many segments; how the field was trained on them; what was refreshed and how often
“HCPs Segmentation on Predicted Brand Growth” (PMSA webinar)	Champions / Loyalists / Churners from a predicted-growth model; messaging per segment	Notebook §9 is this pattern. Note that it is supervised, and ask what the feature window was
“A Method for Physician Segmentation and Brand Activation Prediction Using Claims Data” (PMSA conference)	Method paper connecting claims-derived features to a segmentation pipeline	Which of tonight's four families they build from claims alone; how they handle small denominators
JPMSA Vol. 5, Article 4 — “Leveraging Predictive Analytics to Derive Patient Insights”	Patient-side segmentation, structurally clean	The patient features; how segments were validated; whether governance is discussed at all (it usually was not, then)

Reflection E. The PMSA growth-segmentation webinar labels HCPs Champions / Loyalists / Churners. Using tonight's cross-tab (notebook §9), explain why a 'Churner' in the access-constrained segment and a 'Churner' in the entrenched segment need different plays — and what that says about using a supervised score without a behavioral segmentation underneath it.

Bring to Week 5

- One answer to Reflection B or E, in writing (five sentences).

- Your five segment names from the exercise, and one HCP from each — the NBA engine will decide what to do with them.
- Homework 1 (MMM) is due Monday, October 12, 11:59 PM ET.